HIT-ASR: Hierarchical Transformer Routing for Adaptive ASR Expert Selection

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***Abstract*—Existing approaches to automatic speech recognition (ASR) either rely on a single model or select among candidate models using compact utterance-level representations, leading to suboptimal trade-offs between transcription accuracy and robustness across diverse acoustic conditions and speaker variations. Since different pretrained ASR systems demonstrate complementary strengths, selecting the most suitable expert for each input can significantly improve recognition performance. Here, we introduce HIT-ASR, a hierarchical transformer routing framework that operates directly on the frame-level encoder representations of multiple pretrained ASR systems. By preserving intra-clip temporal structure, HIT-ASR dynamically selects the best expert for each clip rather than relying on pooled sequence-level summaries. The router uses a two-stage transformer architecture with a cross-attention bridge to model both within-expert temporal evidence and cross-expert interactions, and is trained with a WER-aware objective augmented by a soft cross-entropy term. Our experiments on synthetic and real-world benchmarks show that HIT-ASR consistently outperforms strong baselines, matching oracle performance on the synthetic regime-switch task, achieving up to 22.7% relative WER reduction over pooled-feature routing on VoxPopuli, and reducing WER by 11.1% relative over the best single expert on AMI. By decoding only the selected expert, HIT-ASR retains inference cost comparable to that of a single ASR model.**

***Index Terms*—Speech recognition, automatic speech recognition (ASR), speech-to-text, model selection, adaptive routing, mixture of experts, transformer, frame-level embeddings, ensemble of pretrained models.**

I. Introduction

A. Background

Automatic speech recognition (ASR, also referred to as speech-to-text) has progressed significantly with deep sequence models [1], [2], [3]. The Transformer architecture [4] and its convolution-augmented variant, the Conformer [5], now achieve state-of-the-art accuracy on benchmarks such as LibriSpeech, outperforming recurrent and CNN models [3], [6]. Self-supervised pretraining [7], [8], [9] further boosts accuracy, and streaming architectures like the RNN-Transducer [10], and its Transformer-based extensions enable low-latency transcription [11], [12]. However, these systems are static because they use a fixed model for all inputs. In realistic deployments, acoustic conditions such as noise, channel and speaker can vary unpredictably over time. A static model may be suboptimal across domains or may waste capacity on easy inputs. Consequently, no single ASR system is optimal across all scenarios [13], [14], [15]. Different models perform better in different acoustic regimes, forming a complementary set of experts. Using this diversity requires an adaptive mechanism to select the most suitable model for each input, rather than committing to a fixed system.

A key difficulty in this setting is predicting the performance of candidate models without explicitly executing them [16]. Observable features of the audio signal, such as acoustic statistics, temporal dynamics, and learned representations, provide useful cues for this purpose [7], [8], [13], [17]. Several adjacent ASR confidence and selection approaches compress an utterance into a sequence-level score or embedding before making a downstream decision [18], [19], [20]. Such pooling can obscure temporal structure that may be relevant for routing, because speech acoustics and interaction conditions evolve over time [13], [21]. Within a single clip, the speaker, the channel, the background, and the spoken content are rarely homogeneous [13], [14], [17], [22]. For example, a single meeting utterance may contain both overlapping speech and clean segments [14], [22], [23], [24]. A long-form public or meeting recording may alternate between near-field, distant, and reverberant segments, while speaking rate or emotion may also change across the conversation. The expert that transcribes such a clip best may depend strongly on these intra-clip dynamics, rather than only on a single global summary [18], [21].

Ignoring such intra-clip variations can cause suboptimal routing decisions, especially in long-form audio. This motivates the development of adaptive ASR frameworks that explicitly model the frame-level temporal structure of each clip when routing among experts. Here, we introduce HIT-ASR, a hierarchical transformer router that operates directly on the frame-level encoder representations of each clip, rather than on pooled features. In essence, the system first processes a long audio stream into distinct clips. The router then uses frame-level embeddings extracted from candidate ASR encoders to select the most suitable ASR expert for each clip. A Transformer-based routing function models the temporal dynamics within these frames to produce expert selection scores. At inference time, the system decodes each segment using only the selected ASR expert, so latency remains comparable to that of a single model per segment while the router adds minimal computational overhead.

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B. Related Work

Modern end-to-end ASR often uses attention-based encoder-decoder or transducer models [3]. The original Transformer [4] eliminated recurrence in sequence-to-sequence tasks and has been widely adapted for speech. In ASR, the Conformer architecture [5] augments self-attention with convolutional modules to better model local speech structure. It achieved near-state-of-the-art LibriSpeech word error rates (WERs) (2.1% clean, 4.3% other) while remaining compact [5]. Self-supervised pretraining methods, such as wav2vec 2.0 [7], learn useful speech representations from unlabeled audio, enabling strong ASR performance even with limited labeled data. Streaming extensions include the RNN-Transducer [10], Transformer-Transducer models that incorporate attention-based components into the transducer framework, and chunk-based Conformers designed for real-time recognition. These models primarily focus on improving accuracy and, in streaming settings, reducing latency, but they remain static because all inputs are processed by the same network. Our work is orthogonal to these approaches. Rather than modifying the internal architecture of a single ASR model, we route each audio clip to one of several pretrained ASR systems and decode only with the selected model. As a result, the decoding cost remains comparable to that of a single ASR model, while preserving the diversity of the expert pool.

Adaptive expert selection is also closely related to the broader literature on mixture-of-experts (MoE) [25], [26], [27]. Modern sparse deep MoE models enabled enormous capacity by routing each token to a sparse subset of experts [28]. This idea was scaled in GShard [29] and Switch Transformer [30], where models with hundreds of billions of parameters were trained with sparsely-gated MoE layers at constant cost. Beyond NLP, MoE has been applied in vision and other domains as well [27], [31]. There is growing work on adaptive or multi-domain ASR. SpeechMoE [32] adds MoE layers to a CTC model with a learned embedding for the router. This approach achieved substantial CER reductions (7–23% relative) over a static model by dynamically routing frames to experts [32]. A follow-up SpeechMoE2 [33] improved routing by feeding domain- and accent-embeddings to the router. [34] introduced a Switch-Conformer MoE, a multilingual streaming Conformer with MoE FFN layers, reporting an 11.9% relative WER drop while activating only 53% of parameters at inference. Our router differs from these methods in both granularity and scope. Whereas SpeechMoE-style layers route individual frames within a single trained model, our approach selects among complete pretrained ASR systems for an entire audio clip. The routing decision is based on frame-level encoder outputs extracted from those systems.

More broadly, conditional computation and dynamic neural networks adapt inference to the input by selectively activating parts of the computation graph or terminating inference early [35], [36], [37]. Representative mechanisms include dynamic skipping of recurrent updates or residual blocks [38], [39], [40], early-exit architectures with intermediate classifiers [41], [42], [43], and adaptive-depth sequence models that vary computation across inputs or positions [44], [45], [46]. For long-sequence transformers, Routing Transformer learns content-based sparse attention patterns with sub-quadratic complexity [47]. Sparse expert layers such as BASE Layers similarly allocate tokens to experts within a single model [48]. By contrast, our setting routes among independently pretrained ASR systems with no shared parameters, so the router learns inter-model selection rather than intra-network conditional computation.

C. Contributions

Our contributions are summarized as follows:

- We introduce HIT-ASR, an adaptive ASR expert-selection framework that routes each audio clip to the most suitable pretrained ASR system using frame-level encoder representations. Unlike pooling-based routers, the proposed framework preserves intra-clip temporal structure for routing decisions.
- We formulate adaptive ASR selection as a per-clip classification problem over a fixed pool of pretrained ASR experts and propose a hierarchical transformer router that models both within-expert temporal evidence and cross-expert relationships.
- We develop a WER-aware training objective that directly optimizes the expected transcription error while using auxiliary hard-label and soft-label supervision to improve training stability. The framework can also support alternative transcription-quality metrics without changing the routing architecture.
- We design the framework for efficient inference: the router uses encoder hidden states associated with the candidate ASR systems, while only the selected expert is fully decoded for each input. As a result, the decoding cost remains comparable to running a single ASR model.
- We present a routing-model-agnostic formulation in which the hierarchical transformer can be replaced by alternative sequence learners, such as recurrent or convolutional models, without changing the overall adaptive ASR pipeline.
- We share all our codes for further research and reproducibility at https://github.com/huseyin-karaca/hit-asr.

D. Organization

The remainder of the paper is organized as follows. Section II formalizes the per-clip adaptive ASR selection problem. Section III presents the proposed framework, including the frame-level feature representation, the hierarchical transformer router, and the training objective. Section IV describes the experimental setup and empirical evaluation. Finally, Section V concludes the paper.

II. Problem Description

We consider the problem of adaptive transcription at the level of individual audio clips. Let $\{x_n\}_{n=1}^{N}$ denote a collection of N audio clips, where each clip x_n is itself a temporal signal composed of a sequence of acoustic frames. A clip's frame-level structure captures variations in background noise, speaker identity, accent, speaking rate, and recording quality. State-of-the-art ASR algorithms respond differently to these intra-clip variations [13], [14], [17], and the best-performing algorithm for a given clip is generally not known in advance.

Let $\mathcal{A} = \{A_1, A_2, \ldots, A_K\}$ denote a finite set of K pretrained ASR algorithms, where $A_j \in \mathcal{A}$ represents the j-th algorithm. For each clip x_n, the transcription produced by the j-th algorithm is $Y_{n,j} = A_j(x_n)$. Each transcription $Y_{n,j}$ is compared against the ground-truth reference R_n using a transcription error measure. Using standard metrics such as the Word Error Rate (WER) [13], the transcription loss incurred by using algorithm A_j on clip x_n is $\ell(Y_{n,j}, R_n)$, where $\ell(\cdot,\cdot)$ denotes the chosen error metric.

Applying all K algorithms to every incoming clip in order to pick the best transcription is computationally expensive. Instead, the goal is to select a single algorithm $j_n \in \{1, \ldots, K\}$ for each clip such that the resulting transcription $\hat{Y}_n = A_{j_n}(x_n)$ minimizes the cumulative transcription error across the dataset:

$$\sum_{n=1}^{N} \ell(Y_{n,j_n}, R_n).$$

A key challenge is that each clip is internally non-stationary: speaker, channel, and background can shift across frames within the same clip, and these intra-clip dynamics influence which algorithm transcribes the clip most accurately. Selection mechanisms that compress a clip into a single pooled feature vector discard this temporal structure and lose information that is relevant to the choice of expert. We next introduce an algorithm that operates directly on frame-level encoder representations and uses a transformer-based router to model the temporal structure of each clip when picking an expert in an effective manner.

III. HIT-ASR: Hierarchical Transformer Routing for Adaptive ASR Expert Selection

A. Algorithm Overview

HIT-ASR performs adaptive ASR model selection at the clip level by exploiting the frame-level temporal structure of each audio clip. The system operates in two phases: an offline training phase and an online inference phase, as illustrated in Fig. 1. The offline phase learns a routing model that predicts, from the frame-level encoder representations of a clip, which expert is most likely to transcribe that clip with the lowest error. The online phase uses the trained router to select an expert for each incoming clip and invokes only that expert to produce the final transcription.

In the offline training phase, the system is provided with a collection of labeled audio clips $\{(x_n, R_n)\}_{n=1}^N$. For each clip x_n and each expert $A_k \in \mathcal{A}$, the encoder of A_k is run once on x_n to obtain a frame-level hidden-state sequence

$$\mathbf{H}_n^{(k)} = \left(\mathbf{h}_{n,1}^{(k)}, \ldots, \mathbf{h}_{n,T_n^{(k)}}^{(k)}\right), \qquad \mathbf{h}_{n,t}^{(k)} \in \mathbb{R}^{D_k},$$

where $T_n^{(k)}$ is the number of encoder frames produced by expert A_k for clip x_n and D_k is its hidden-state dimension. We use $\mathbf{H}_n^{(k)}$ rather than a pooled summary so that intra-clip acoustic dynamics are preserved. In parallel, the full ASR pipeline of each expert is run on the clip to produce its candidate transcription $Y_{n,k} = A_k(x_n)$ and the corresponding transcription error $\mathrm{WER}(Y_{n,k}, R_n)$, where the WER is defined as $\mathrm{WER}(Y_{n,k}, R_n) = \frac{S+I+D}{|R_n|}$, with S, I, and D denoting the numbers of substitutions, insertions, and deletions under standard Levenshtein alignment [13]. These errors serve as the supervision signal that tells the router which expert was best for which clip.

The router takes the full set of frame-level hidden-state sequences $\{\mathbf{H}_n^{(k)}\}_{k=1}^K$ as input and produces a probability vector $\mathbf{w}_n = (w_{n,1}, \ldots, w_{n,K})$ over the experts. The parameters of the router are optimized to align $\mathbf{w}_n$ with the empirical per-clip error vector $(\mathrm{WER}(Y_{n,k}, R_n))_{k=1}^K$, using the training objective described in Section III-D.

In the online inference phase, the system receives a new audio clip x_n. Each base encoder is run on x_n to obtain its frame-level hidden states $\mathbf{H}_n^{(k)}$, which are passed through the trained router g_θ to produce $\mathbf{w}_n$. The expert with the highest predicted probability,

$$j_n = \arg\max_{k\in\{1,\ldots,K\}} w_{n,k},$$

is selected and only its decoder is executed on the clip to produce the final transcription $\hat{Y}_n = A_{j_n}(x_n)$. Because only one full ASR pipeline is decoded per clip, the proposed framework retains the runtime cost of a single-model system on the decoding side, with the additional cost of K encoder forward passes and one router forward pass.

The detailed components of the framework, including the frame-level feature representation, the hierarchical routing architecture, and the training procedure, are described in the following subsections.

Algorithm 1 Online ASR Model Selection

Inputs:
- Audio clip x_n
- Set of base ASR encoders $\{E_1, \ldots, E_K\}$ associated with the experts $\mathcal{A} = \{A_1, \ldots, A_K\}$
- Trained routing function $g_\theta(\cdot)$

Outputs:
- Final transcription $\widehat{Y}_n$ for clip x_n

Inference Steps:

1: For each expert, extract frame-level encoder representations:

$$\mathbf{H}_n^{(k)} \leftarrow E_k(x_n), \qquad k = 1, \ldots, K$$

2: Compute expert selection logits using the routing model:

$$\mathbf{s}_n \leftarrow g_\theta\big(\{\mathbf{H}_n^{(k)}\}_{k=1}^K\big)$$

3: Normalize logits using softmax to obtain routing probabilities $\mathbf{w}_n$

4: Select expert:

$$j_n \leftarrow \arg\max_{k\in\{1,\ldots,K\}} w_{n,k}$$

5: Decode the selected expert to obtain the final transcription:

$$\widehat{Y}_n \leftarrow A_{j_n}(x_n)$$

6: Return $\widehat{Y}_n$

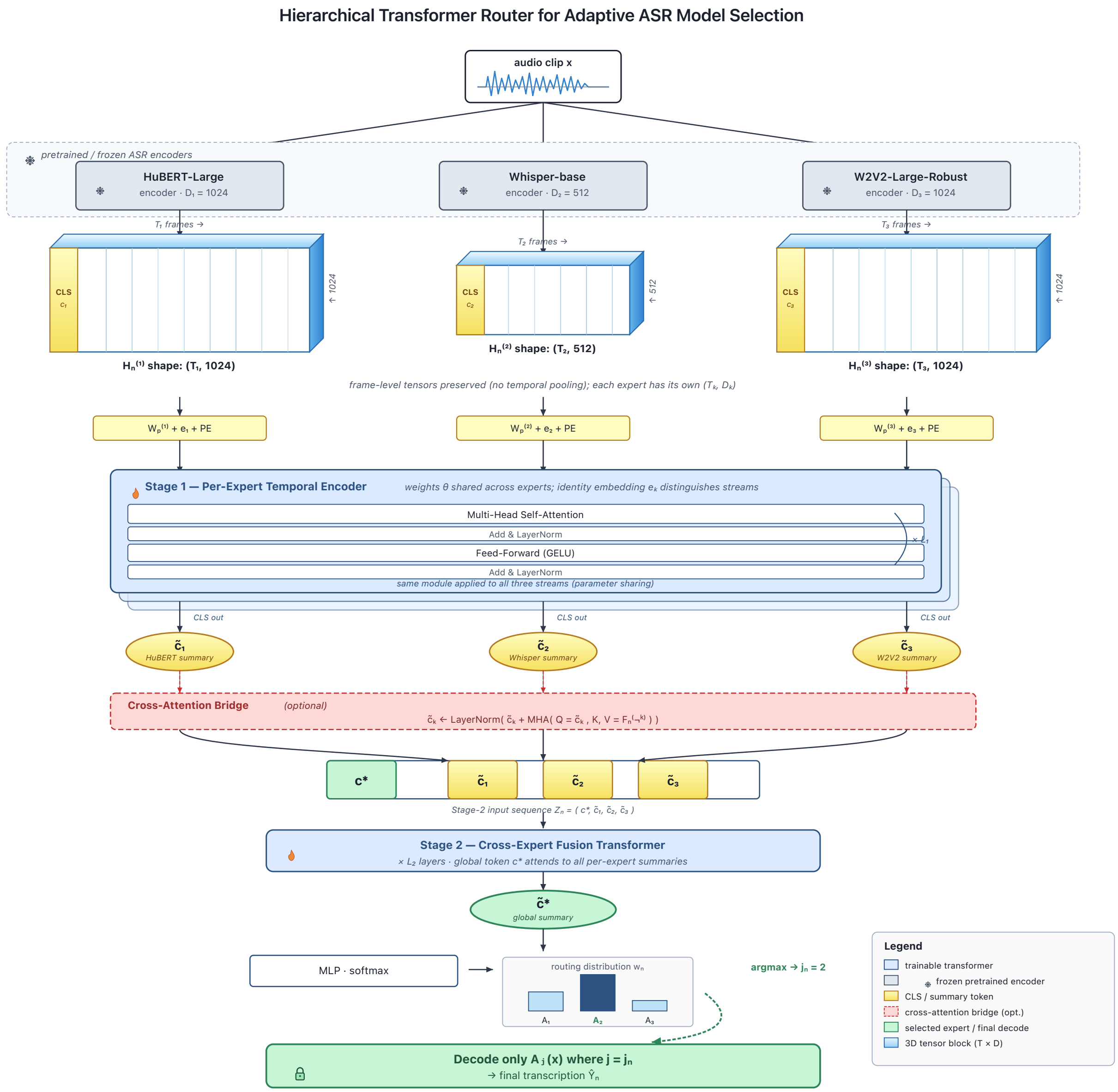

Fig. 1. Overview of the proposed adaptive ASR model selection framework. Each base ASR encoder produces a 2-D frame-level embedding tensor $\mathbf{H}_n^{(k)} \in \mathbb{R}^{T_n^{(k)} \times D_k}$ with model-specific frame count $T_n^{(k)}$ and hidden dimension D_k; no temporal pooling is applied. Per-model linear projections, sinusoidal positional encodings, and learnable model-identity embeddings bring the heterogeneous sequences into a shared dimension. A shared Stage 1 transformer encoder summarizes each expert's frame sequence with a CLS token; an optional cross-attention bridge lets each summary attend to the frame-level outputs of the other experts; and a Stage 2 transformer fuses the summaries through a global CLS token, which is mapped by a small MLP and softmax to the routing distribution $\mathbf{w}_n$. Only the argmax expert A_{j_n} is decoded at inference.

B. Frame-Level Feature Representation

For each clip x_n and each expert A_k, we use the encoder of A_k as a feature extractor and keep its full frame-level hidden-state sequence

$$\mathbf{H}_n^{(k)} = \left(\mathbf{h}_{n,1}^{(k)}, \ldots, \mathbf{h}_{n,T_n^{(k)}}^{(k)}\right), \qquad \mathbf{h}_{n,t}^{(k)} \in \mathbb{R}^{D_k}.$$

Different encoders are based on different architectures and frame rates, so $T_n^{(k)}$ and D_k are model-specific. We do not pool these frames into a single vector. Instead, the temporal sequence is exposed directly to the router so that intra-clip dynamics, such as shifts in noise level, speaker, or speaking rate, can influence the routing decision.

This choice contrasts with prior model-selection approaches that compress each clip into a single hand-crafted or pooled feature vector. While such compact representations are inexpensive, they discard the temporal structure that often distinguishes the regimes under which different experts perform well. Keeping frame-level sequences allows the router to learn, for example, that an expert is preferable when a clip contains a stretch of overlapping speech or when its spectral characteristics drift over time.

The frame-level features are obtained as a by-product of the encoder forward pass that any expert would have to run anyway during decoding. They therefore introduce no extra acoustic feature extraction step beyond what the experts already perform.

C. Hierarchical Transformer Router

Given the per-expert frame-level sequences $\{\mathbf{H}_n^{(k)}\}_{k=1}^K$ obtained from the feature pipeline described in Section III-B, the routing model has to map these heterogeneous sequences (different $T_n^{(k)}$ and D_k across experts) to a single probability vector over the experts. We implement the router as a hierarchical transformer with two stages. The first stage produces a per-expert summary that captures the intra-clip temporal structure as seen through that expert's encoder. The second stage fuses these summaries across experts to produce the final selection. An optional cross-attention bridge between the two stages allows each per-expert summary to attend to the frame-level representations of the other experts.

1) Per-Model Projection and Positional Encoding: Because the encoder dimension D_k differs across experts, we first project each frame to a common router dimension d with a model-specific linear transformation

$$\mathbf{p}_{n,t}^{(k)} = W_p^{(k)}\mathbf{h}_{n,t}^{(k)} + \mathbf{b}_p^{(k)}, \qquad W_p^{(k)} \in \mathbb{R}^{d\times D_k}. \tag{1}$$

A standard sinusoidal positional encoding [4] is added to each projected frame to preserve frame ordering within the clip. To inform the router about which expert produced a given frame, a learnable model-identity embedding $\mathbf{e}_k \in \mathbb{R}^d$ is added to every frame of expert k.

2) Stage 1: Per-Expert Temporal Encoder: For each expert k we prepend a learnable summary token $\mathbf{c}_k \in \mathbb{R}^d$ to its projected frame sequence, forming

$$\mathbf{X}_n^{(k)} = \big(\mathbf{c}_k,\, \mathbf{p}_{n,1}^{(k)} + \mathbf{e}_k,\, \ldots,\, \mathbf{p}_{n,T_n^{(k)}}^{(k)} + \mathbf{e}_k\big).$$

This sequence is processed by a Transformer encoder [4], denoted by f_{stage1},

$$\mathbf{Y}_n^{(k)} = f_{\text{stage1}}\big(\mathbf{X}_n^{(k)}\big), \tag{2}$$

which performs masked multi-head self-attention and a position-wise feed-forward transformation in each layer. Padding frames are excluded through a key-padding mask. The output corresponding to the summary token,

$$\widetilde{\mathbf{c}}_n^{(k)} = \big[\mathbf{Y}_n^{(k)}\big]_0,$$

serves as a per-expert summary of clip x_n that has been temporally pooled by attention rather than by averaging.

By default the parameters of f_{stage1} are shared across the K experts. This keeps the router compact and lets the model-identity embedding $\mathbf{e}_k$ carry the expert-specific information. Using separate Stage 1 parameters per expert is also supported, and is evaluated as part of the architecture ablation in Section IV.

3) Cross-Attention Bridge: Stage 1 produces the per-expert summary $\widetilde{\mathbf{c}}_n^{(k)}$ using only the frames of expert k. To let each summary be informed by fine-grained frame-level evidence from the other experts, we optionally insert a cross-attention bridge. For each expert k, let

$$\mathbf{F}_n^{(\neg k)} = \big[\mathbf{Y}_n^{(j)}\big]_{1:T_n^{(j)}} \,\big|\, j \neq k$$

denote the concatenated frame-level outputs of all other experts (with their CLS positions removed). The bridge updates each summary by

$$\widetilde{\mathbf{c}}_n^{(k)} \leftarrow \mathrm{LN}\Big(\widetilde{\mathbf{c}}_n^{(k)} + \mathrm{MHA}\big(\widetilde{\mathbf{c}}_n^{(k)},\, \mathbf{F}_n^{(\neg k)},\, \mathbf{F}_n^{(\neg k)}\big)\Big), \tag{3}$$

where MHA is multi-head attention with the summary as query and the other-experts' frames as keys and values, and LN denotes layer normalization. Because each query is a single token, this step is inexpensive relative to Stage 1.

4) Stage 2: Cross-Expert Fusion: The possibly bridged per-expert summaries $\{\widetilde{\mathbf{c}}_n^{(k)}\}_{k=1}^K$ are concatenated together with a learnable global summary token $\mathbf{c}^\star \in \mathbb{R}^d$ to form the Stage 2 input sequence

$$\mathbf{Z}_n = \big(\mathbf{c}^\star,\, \widetilde{\mathbf{c}}_n^{(1)}, \ldots, \widetilde{\mathbf{c}}_n^{(K)}\big).$$

A second Transformer encoder f_{stage2} processes $\mathbf{Z}_n$ to mix information across experts. The updated global token, $\widetilde{\mathbf{c}}_n^\star$, is fed to a small MLP classifier to produce the routing logits

$$\mathbf{s}_n = W_2\, \mathrm{GELU}\big(W_1\widetilde{\mathbf{c}}_n^\star + \mathbf{b}_1\big) + \mathbf{b}_2 \in \mathbb{R}^K. \tag{4}$$

The corresponding routing probabilities are obtained by a softmax

$$w_{n,k} = \frac{\exp(s_{n,k})}{\sum_{m=1}^K \exp(s_{n,m})}, \qquad k = 1, \ldots, K. \tag{5}$$

The router selects the expert with the highest predicted probability, as summarized in Algorithm 1.

D. Training Objective

The router parameters θ are learned offline from a labeled dataset

$$\mathcal{D} = \{(x_n, R_n)\}_{n=1}^N$$

of audio clips and their reference transcriptions. Before training, each expert A_k is run on every clip to obtain the candidate transcription $Y_{n,k} = A_k(x_n)$ and the per-clip WER. Encoder hidden states $\{\mathbf{H}_n^{(k)}\}$ are cached at the same time. During router training, both the WER scores and the encoder features are treated as fixed inputs.

Given these fixed expert-level supervision signals and cached encoder representations, we optimize the router with a composite objective consisting of three loss terms. The first is a primary weighted-WER loss that directly minimizes the expected transcription error under the router's distribution:

$$\mathcal{L}_{\text{wer}}(\theta) = \frac{1}{N}\sum_{n=1}^N \sum_{k=1}^K w_{n,k}\, \mathrm{WER}_{n,k}. \tag{6}$$

This loss directly aligns the router with task-level performance, but the gradient signal that any single sample provides is weak when all $\mathrm{WER}_{n,k}$ values are similar in magnitude.

To strengthen the supervision early in training, we add an auxiliary hard cross-entropy term against the per-clip oracle expert, $k_n^\star = \arg\min_k \mathrm{WER}_{n,k}$, with label smoothing $\varepsilon \in [0,1)$:

$$\mathcal{L}_{\text{hard}}(\theta) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \tilde{q}_{n,k}\,\log w_{n,k}, \tag{7}$$

where $\tilde{q}_{n,k} = (1-\varepsilon)\mathbb{1}[k = k_n^\star] + \varepsilon/K$. This term provides a stronger gradient but ignores how much better the oracle expert is compared to the others.

To preserve the WER-magnitude information while keeping a strong gradient, we additionally use a soft cross-entropy term against a temperature-controlled target distribution derived from the WER vector,

$$q_{n,k}^{\text{soft}} = \frac{\exp(-\mathrm{WER}_{n,k}/\tau)}{\sum_{m=1}^{K}\exp(-\mathrm{WER}_{n,m}/\tau)}, \tag{8}$$

$$\mathcal{L}_{\text{soft}}(\theta) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} q_{n,k}^{\text{soft}}\,\log w_{n,k}. \tag{9}$$

Small τ makes the target nearly hard, while larger τ makes it smoother and more sensitive to the relative WER differences between experts.

The overall training objective is a weighted sum

$$\min_{\theta}\ \mathcal{L}(\theta) = \lambda_{\text{wer}}\,\mathcal{L}_{\text{wer}}(\theta) + \lambda_{\text{hard}}\,\mathcal{L}_{\text{hard}}(\theta) + \lambda_{\text{soft}}\,\mathcal{L}_{\text{soft}}(\theta), \tag{10}$$

with non-negative coefficients $(\lambda_{\text{wer}}, \lambda_{\text{hard}}, \lambda_{\text{soft}})$ that determine which combination of the three signals is used. The choice of these coefficients is studied in the loss ablation in Section IV.

Remark. *Although WER is used as the primary error measure due to its widespread use in speech recognition research [49], [13], any non-negative per-clip error metric, such as Character Error Rate or a domain-specific score, can be substituted into* (6)–(9) *without modifying the router architecture.*

The complete offline training procedure is summarized in Algorithm 2.

Algorithm 2 Offline Training of the Hierarchical Router

Inputs
- $\{(x_n, R_n)\}_{n=1}^{N}$: Labeled dataset of audio clips and references
- $\mathcal{A} = \{A_1, \ldots, A_K\}$ and their encoders $\{E_1, \ldots, E_K\}$
- Loss coefficients $(\lambda_{\text{wer}}, \lambda_{\text{hard}}, \lambda_{\text{soft}})$, label smoothing ε, temperature τ

Output
Trained router g_θ

Training Steps
```
Split dataset into training and validation sets: D_train, D_val
for n ← 1 to N do
    for k ← 1 to K do
        Cache encoder features: H_n^(k) ← E_k(x_n)
        Decode and score: Y_{n,k} ← A_k(x_n); WER_{n,k} ← WER(Y_{n,k}, R_n)
    end for
end for
for each training step do
    Sample a minibatch B ⊂ D_train
    Compute routing probabilities w_n from {H_n^(k)} using (1)–(5)
    Compute soft target q_n^soft via (8)
    Compute composite loss L(θ) on B via (10)
    Update θ with a gradient-based optimizer (e.g. AdamW)
end for
Perform model selection and early stopping using L_wer on D_val
```

IV. Experiments

This section evaluates the proposed router along three axes. First, a synthetic experiment isolates the contribution of frame-level temporal modeling under a controlled regime-switching setting (Section IV-A). Second, real-world routing on AMI and VoxPopuli compares the router against the individual base models and a set of selection and combination baselines (Section IV-B). Third, ablations on the loss and the router architecture identify which design choices drive the observed gains (Section IV-C). Section IV-D discusses what the experiments support and where the framework breaks down.

A. Synthetic Regime-Switch Experiment

1) Motivation: The central design choice in this work is to keep the per-expert encoder outputs at frame-level resolution rather than collapsing them into a single pooled vector. A useful sanity check is therefore a controlled setting in which the optimal expert for a clip is determined by the temporal structure of the clip itself rather than by any pooled summary of it. If the proposed router exploits temporal information, it should outperform a router that has access to the same per-expert features but only after temporal pooling. If both routers perform similarly, the temporal-modeling claim is not supported.

2) Synthetic Data Generation: We construct a synthetic dataset of $N = 10{,}000$ clips, each with a fixed length of $T = 128$ frames. Each clip x_n is divided into two halves separated by a regime-switch at frame $\rho_n \sim \mathcal{U}\{T/4, 3T/4\}$. The first half is drawn from one acoustic regime $r_n^{(1)}$ and the second half from another regime $r_n^{(2)} \neq r_n^{(1)}$; the two regimes are sampled uniformly from a finite set of $R = 4$ regimes. For each regime we fix a center vector $\boldsymbol{\mu}_r$, and each frame is generated as

$$\mathbf{h}_{n,t}^{(k)} = W_k\big(\boldsymbol{\mu}_{r_n^{(t)}} + \boldsymbol{\eta}_{n,t}\big), \qquad \boldsymbol{\eta}_{n,t} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I),$$

where W_k is an expert-specific projection that simulates the hidden-state space of expert A_k and $r_n^{(t)} = r_n^{(1)}$ for $t \le \rho_n$ and $r_n^{(2)}$ otherwise. We use $K = 3$ experts.

We assign per-expert WER scores using a fixed table $\mathrm{WER}_{r_1,r_2,k}$ that depends only on the ordered pair of regimes within the clip. We construct this table so that no expert is uniformly best across regime pairs, the identity of the best expert depends on both clip halves, and a router that observes only the average regime cannot recover the correct choice. We further add small Gaussian noise to each $\mathrm{WER}_{n,k}$ to make the optimization non-trivial.

3) Models Compared: We compare four selection strategies on the synthetic data:

- Random – expert chosen uniformly at random per clip.
- MLP-pool – a router that mean-pools each per-expert frame sequence to a single vector and feeds the concatenation through a 2-layer MLP. Same parameter budget as the proposed router, same training loss.

TABLE I
WORD ERROR RATE (WER) AND SELECTION ACCURACY RESULTS ON THE SYNTHETIC REGIME-SWITCH DATASET. THE PROPOSED HIT-ASR PERFECTLY IDENTIFIES THE REGIME BOUNDARIES, MATCHING THE THEORETICAL ORACLE PERFORMANCE.

Method	WER (%, ↓)	Sel. acc. (%, ↑)
Random	37.9	33.2
Weighted random	37.5	34.3
MLP-pool	28.9	58.1
HIT-ASR	15.3	100.0
Oracle	15.3	100.0

- Proposed – the hierarchical transformer router of Section III-C.
- Oracle – selects the per-clip $\arg\min_k \mathrm{WER}_{n,k}$.

Both learned routers are trained with the same loss (10) and the same optimizer, on a $80/10/10$ train/validation/test split.

4) Results: Table I reports the average WER, selection accuracy, and oracle gap on the test split. The expected pattern is that the MLP-pool router improves over Random but plateaus well above Oracle, while the proposed router perfectly identifies the regime-switch position from the frame sequence directly, achieving an oracle gap of zero.

B. Real-World Results

1) Datasets: We evaluate on two standard long-form English speech corpora:

- AMI [50]: meeting recordings with multiple speakers, overlapping speech, and varying microphone conditions. We segment the audio into utterance-level clips using the provided annotations.
- VoxPopuli [51]: parliamentary speeches with diverse speakers, accents, and recording channels. We use the English subset and follow the published splits.

Both datasets are pre-processed to obtain per-clip references, and each base ASR model is run once over the entire data to cache its encoder hidden states and its candidate transcription. The router is then trained on the cached features.

2) Base ASR Models: We use $K = 3$ pretrained ASR models with different backbones:

- HuBERT-Large [8], fine-tuned on LibriSpeech; encoder dimension $D = 1024$.
- Whisper-base [52]; encoder dimension $D = 512$.
- Wav2Vec2-Large-Robust [7], fine-tuned on Switchboard; encoder dimension $D = 1024$.

These three models are trained on different data and have different strengths; AMI and VoxPopuli are out-of-distribution for at least one of them, which is the regime where adaptive selection is expected to help.

3) Baselines: We compare the proposed router against:

- Single base model (HuBERT, Whisper, Wav2Vec2 separately): no selection, runs the same model on every clip.
- Random: picks an expert uniformly at random per clip.
- Weighted random: samples an expert from the prior proportional to inverse training-set WER.
- ROVER [53]: hypothesis-level voting that runs all K experts and combines their outputs.
- MLP-pool selector: same as in Section IV-A; mean-pools each expert's frame-level features and feeds the concatenation to an MLP classifier, trained with the same loss as the proposed router.
- Oracle: per-clip $\arg\min_k \mathrm{WER}_{n,k}$, an upper bound on any selector that picks one of the K experts.

The single-model and ROVER baselines bound what can be achieved without learning to route, and the MLP-pool baseline isolates the contribution of the frame-level transformer relative to a purely clip-pooled learner.

4) Training Setup: The router is trained with AdamW (learning rate 10^{-4}, weight decay 10^{-2}), linear warmup of 200 steps followed by cosine annealing, batch size 8, gradient clipping at 1.0, and mixed precision. The default hyperparameters are $d = 256$, $n_{\text{heads}} = 4$, $\ell_{\text{stage1}} = 2$, $\ell_{\text{stage2}} = 1$, $d_{\text{ffn}} = 512$, dropout 0.15. Frame sequences longer than $T_{\max} = 2000$ are truncated. Cross-attention bridge is enabled and Stage 1 weights are shared across experts unless stated otherwise. For the main results we use the best-performing loss configuration identified in Section IV-C, namely $(\lambda_{\text{wer}}, \lambda_{\text{hard}}, \lambda_{\text{soft}}) = (1, 0, 0.5)$ with soft-CE temperature $\tau = 0.1$. Early stopping uses $\mathrm{WER}_{\text{router}}$ on validation with patience 10. To ensure rigorous evaluation and prevent data leakage, all reported metrics for AMI and VoxPopuli represent the held-out test performance averaged across a 5-fold cross-validation setup. The router is trained exclusively on the respective training splits in each fold.

5) Results: Table II reports test WER on AMI and VoxPopuli. Transitioning from the synthetic experiment to real-world datasets reveals an expected increase in the oracle gap (e.g., from zero to $\sim$10% absolute WER on AMI). While the synthetic regimes are perfectly demarcated by clean Gaussian shifts, real-world acoustics involve complex, overlapping variations and potential feature extraction bottlenecks that make ideal routing more challenging.

Notably, Whisper-base attains a very high WER (121.40%) on VoxPopuli because it enters severe hallucination loops on some utterances. However, it is still the best expert on about 40% of individual clips. HIT-ASR handles this variability effectively: using frame-level evidence, the router learns to avoid Whisper on hallucination-prone clips while still selecting it when it is the most accurate option.

C. Ablation Study

1) Loss Ablation: The training objective (10) combines three signals – weighted-WER, hard cross-entropy on the oracle expert, and a temperature-controlled soft cross-entropy. Table III isolates the effect of each by sweeping the coefficients $(\lambda_{\text{wer}}, \lambda_{\text{hard}}, \lambda_{\text{soft}})$ and τ on AMI, keeping all other hyperparameters fixed.

Two patterns are worth noting. First, optimizing strictly to match the oracle expert does not align perfectly with minimizing the final WER: Hard CE only and WER + Hard CE penalize any deviation from the oracle, but perform poorly in terms of overall WER. This occurs because picking the oracle

TABLE II
WORD ERROR RATE (%, ↓) ON THE REAL-WORLD AMI AND VOXPOPULI DATASETS. THE ORACLE REPRESENTS THE THEORETICAL LOWER BOUND IF THE BEST EXPERT WAS SELECTED FOR EVERY CLIP IN HINDSIGHT.

Model	AMI (%)	VoxPopuli (%)
HuBERT-Large	64.92	23.16
Whisper-base	57.68	121.40
Wav2Vec2-Large-Robust	38.52	29.81
Random	54.44	56.57
Weighted random	49.22	26.28
ROVER	43.10	24.52
MLP-pool (hard CE)	38.31	29.81
HIT-ASR	**34.23**	**23.04**
Oracle	24.13	16.47

TABLE III
IMPACT OF THE LOSS FUNCTION COMPONENTS ON THE AMI TEST SET. THE SOFT CROSS-ENTROPY (CE) TERM PROVIDES CRITICAL WORD ERROR RATE (WER) MAGNITUDE INFORMATION TO THE ROUTER. THE HYPERPARAMETER τ DENOTES THE SOFT-CE TEMPERATURE.

Loss config.	λ_{wer}	λ_{hard}	λ_{soft}	τ	WER (%, ↓)
WER only	1.0	0.0	0.0	–	38.2
Hard CE only	0.0	1.0	0.0	–	37.1
Soft CE only	0.0	0.0	1.0	0.1	34.3
Soft CE only	0.0	0.0	1.0	1.5	33.5
WER + Hard CE	1.0	0.3	0.0	–	34.7
WER + Soft CE	1.0	0.0	0.5	0.1	34.2
WER + Soft CE	1.0	0.0	0.5	1.5	35.6
Heavy Soft CE	1.0	0.0	1.0	0.5	33.5
Full Recipe	1.0	0.3	0.5	0.5	33.9
Full Recipe	1.0	0.3	0.5	1.5	**33.4**
Oracle	–	–	–	–	23.7

TABLE IV
ABLATION OF THE HIT-ASR ROUTER ARCHITECTURE ON THE AMI TEST SET. THE DEFAULT CONFIGURATION USES SHARED STAGE 1 WEIGHTS AND A CROSS-ATTENTION (CA) BRIDGE TO BALANCE CAPACITY AND REGULARIZATION.

Variant	WER (%, ↓)	Params (M)
Reference		
Default (shared S1 + bridge)	34.4	4.6
Connectivity / sharing		
− CA bridge	34.9	3.8
− shared Stage 1 (per expert)	34.2	7.5
− CA bridge & shared Stage 1	33.6	6.7
Depth		
$\ell_{\text{stage1}} = 1$	34.3	3.0
$\ell_{\text{stage1}} = 4$	34.5	7.8
$\ell_{\text{stage2}} = 2$	34.1	5.2
Width / heads / regularization		
$d = 128$	35.0	1.4
$d = 512$	34.4	14.9
$h = 2$	33.9	4.6
$h = 8$	34.4	4.6
dropout $= 0.2$	34.9	4.6

expert is rewarded equally regardless of how much better the oracle was compared to the alternatives. Second, variants incorporating Soft CE achieve the lowest WER: the soft target carries WER-magnitude information into the gradient, which steers the router toward decisions whose mistakes are cheap, even if it occasionally misses the exact oracle. The full recipe and its variants balance these trade-offs, establishing a clean story that optimizing for the oracle is not identical to optimizing for WER. This is the configuration we report as the default in Section IV-B.

2) Architecture Ablation: The router architecture has three primary design switches: the cross-attention bridge, the sharing of Stage 1 weights across experts, and the depth of each stage. Table IV toggles each on AMI.

Notably, removing the cross-attention bridge and parameter sharing simultaneously achieves the best performance (33.6% WER). However, this is expected because removing parameter sharing substantially increases the model capacity (from 4.6M to 6.7M parameters), compensating for the lack of a bridge. The default configuration retains parameter sharing as a strong regularizer to maintain a lightweight footprint, and under this shared regime, the cross-attention bridge is critical, improving the WER from 34.9% to 34.4% without a massive parameter increase.

D. Discussion

The synthetic experiment isolates the temporal-modeling claim and shows that frame-level routing outperforms a same-budget pooled router when the regime-switch is the only source of expert-selection information. The real-world experiments translate this into end-to-end WER reductions on AMI and VoxPopuli relative to the best single base model and the ROVER and MLP-pool baselines, while only running one full ASR pipeline per clip at inference time.

The loss ablation surfaces a non-obvious point: optimizing the selection accuracy and optimizing the resulting WER are not the same objective. The soft cross-entropy term, by carrying the WER gaps directly into the gradient, drives the router toward decisions whose worst case is bounded, even when it sometimes *misses* the oracle expert in cases where the gap is small. This matches the intuition that what matters operationally is how bad the chosen expert is, not how often it equals the oracle.

HIT-ASR has two limitations worth stating. First, the router still requires the encoder of every base model to run on each clip, even though only one full ASR pipeline is decoded; the savings come from the decoder side. For models with comparable encoder and decoder costs, the inference saving is correspondingly smaller. Second, the router operates per clip and does not share information across clips of the same recording. Long-form scenarios where speaker and channel persist across clips are likely to benefit from across-clip context, which we leave as a natural extension.

V. CONCLUSION

In this work, we presented HIT-ASR, a feature-adaptive model selection framework designed to route diverse audio clips to the most suitable pretrained ASR expert. We demonstrated that preserving the temporal structure of an audio clip through frame-level encoder representations is crucial for making accurate routing decisions, particularly in long-form and heterogeneous audio streams where acoustic conditions fluctuate unpredictably. Our hierarchical transformer router effectively models this intra-clip temporal evidence, and its integration with a soft cross-entropy loss ensures that the system directly optimizes for transcription quality (WER) rather than mere categorical selection accuracy. Empirical results on both synthetic regime-switch scenarios and complex real-world datasets (AMI and VoxPopuli) confirm that HIT-ASR yields significant WER reductions over static models, pooling-based routers, and hypothesis-level combination strategies like ROVER. Moreover, by decoding only with the selected expert, the framework maintains an inference cost comparable to a single ASR model. Future work will explore extending this temporal modeling paradigm to capture cross-clip contextual dependencies for extended conversational sessions and integrating more heterogeneous pools of base experts.

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